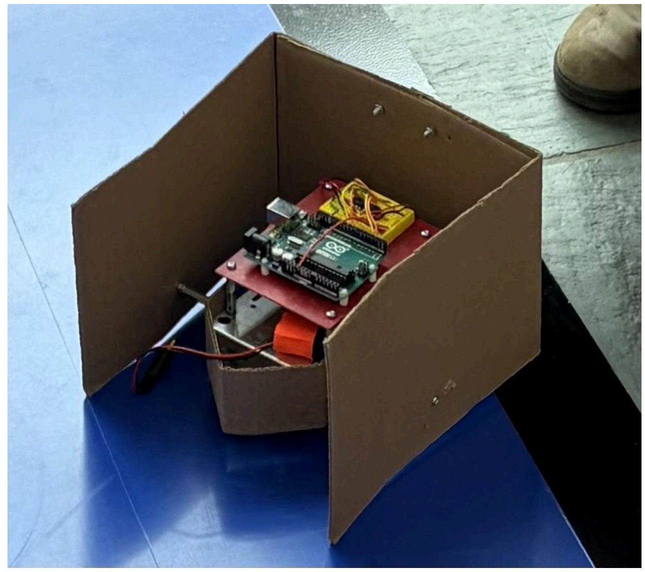
Group 38 - Ella Johnson (emj67), Hannah Mekky (hbm35), Madeline Nason (mln63)

Robot Design and Strategy Overview

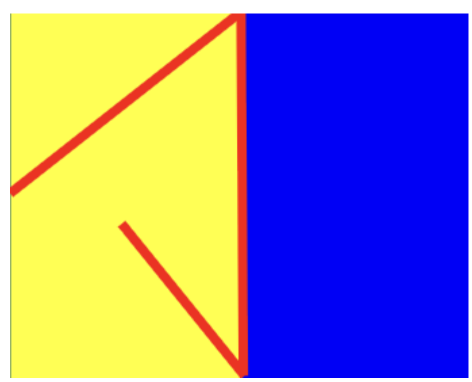


We designed our robot to be as simple as possible. We added three cardboard walls to the robot chassis provided in order to collect the blocks it passes over and by during the competition. Cardboard was added in front of the forward wheel to ensure the blocks would not disrupt the movement of the robot by blocking the wheel or getting stuck under the chassis. The tall walls and wide robot had an added benefit in preventing the robot from being knocked over in the competition when it inevitably ran into other robots, allowing it to keep moving as well as retain its blocks.

Because we chose not to use any sensors or other electrical components besides the two motors used to run the back wheels, our electrical design was fairly simple. Each of the motors was connected to an H-bridge, and the left motor was controlled by pins in PORTD (pins 2 and 3) and the right motor was controlled by pins in PORTB (pins 8 and 9).

We opted not to use sensors of any type in our design. We would have only used the sensors to tell the robot when to switch from driving forward to turning, and instead used delays and measured the amount of time it takes for the robot to drive the specified distance along the path. Our software was designed to have the robot drive to the border between yellow, blue, and the edge of the board, turn, drive along the length of the blue and yellow border, turn again, and drive to the middle of the starting side in order to gather as many blocks along the border as possible. While this design could not account for any external factors such as the robot being set at a different angle than expected or it running into another robot, we chose to hard code a path in due to our limited experience with Arduinos, and trusted that our mechanical design was large enough to not be affected too much by running into other obstacles.

Design Process Reflection



For each milestone, we wanted to be confident in the fundamentals of the robot, and then see if we could expand upon it. For the design pitch in milestone 1, we laid out our thinking for a simple, but efficient method. After constructing the circuit, we wanted to prioritize maneuverability, model supports, and an efficient way of gathering cubes. To the right is the roadmap of our goal route of the bot.

For milestone 2, once the motor wiring with the two H-Bridges was confirmed and we built the base configuration with the arduino and the boe bot chassis, we constructed our code for Milestone 2 by mapping each motor direction to specific pin combinations. We also included a delay to allow for mechanical adjustments and software timing calibration. Making sure each direction worked individually, we then compiled a script to meet the milestone requirements.

Moving onto milestone 3 and building upon milestone 2 along with the code from Lab 4, we implemented while loops along with threshold values for the color sensor to signify when the bot will drive forward and when it will turn to the next direction. The color detection was built upon from the Lab 4 code, and the direction change of the bot came from milestone 2. For example, if starting on yellow, it will loop until the sensor detects blue, turn 180°, drive forward while on yellow, and stop when black is detected. The same occurs for if starting on blue.

Finally for milestone 4, we used the same logic for the color sensing from milestone 3. For the cube collecting aspect, we added a wall design that was designed to have a “collector” concept where the blocks could gather. We also added an additional piece in the front of the bot to guide the blocks to either side of the robot or within the walls themselves. We used standoffs and screws to secure the walls and we placed them in a position to keep the blocks from interfering with the wheels.

For the final bot, after creating the basic layout, we ordered larger wheels and corrugated plastic for additional boosts, but we ran into some challenges as we needed some extra parts to secure the wheels, so we didn’t end up using them. The corrugated plastic had some delivery issues so we couldn’t end up using that either, so we ended up sticking with the cardboard variation of what we used for milestone 4. Besides these logistical roadblocks, what took the most time was debugging the code and making sure that the bot functioned according to each milestone and how we wished for it to work based on our design pitch.

Competition Analysis

Our robot won three out of the six matches it participated in. The large design of the robot was an asset, as it was not affected too much by colliding with other robots, and ended up overturning one. In one of the matches the battery became partly disconnected from the holder, which was not realized until it was too late to fix, causing the robot to be unable to run for the round, which we later fixed using tape. In another round the angle the robot was set at was slightly off, causing the robot to run part way off the board and become stuck. In that case, hardcoding the robot was a weakness, as it could not detect the black border and ran off it. Despite those mishaps, our robot was still relatively successful, and managed to collect an average of 6 blocks per round.

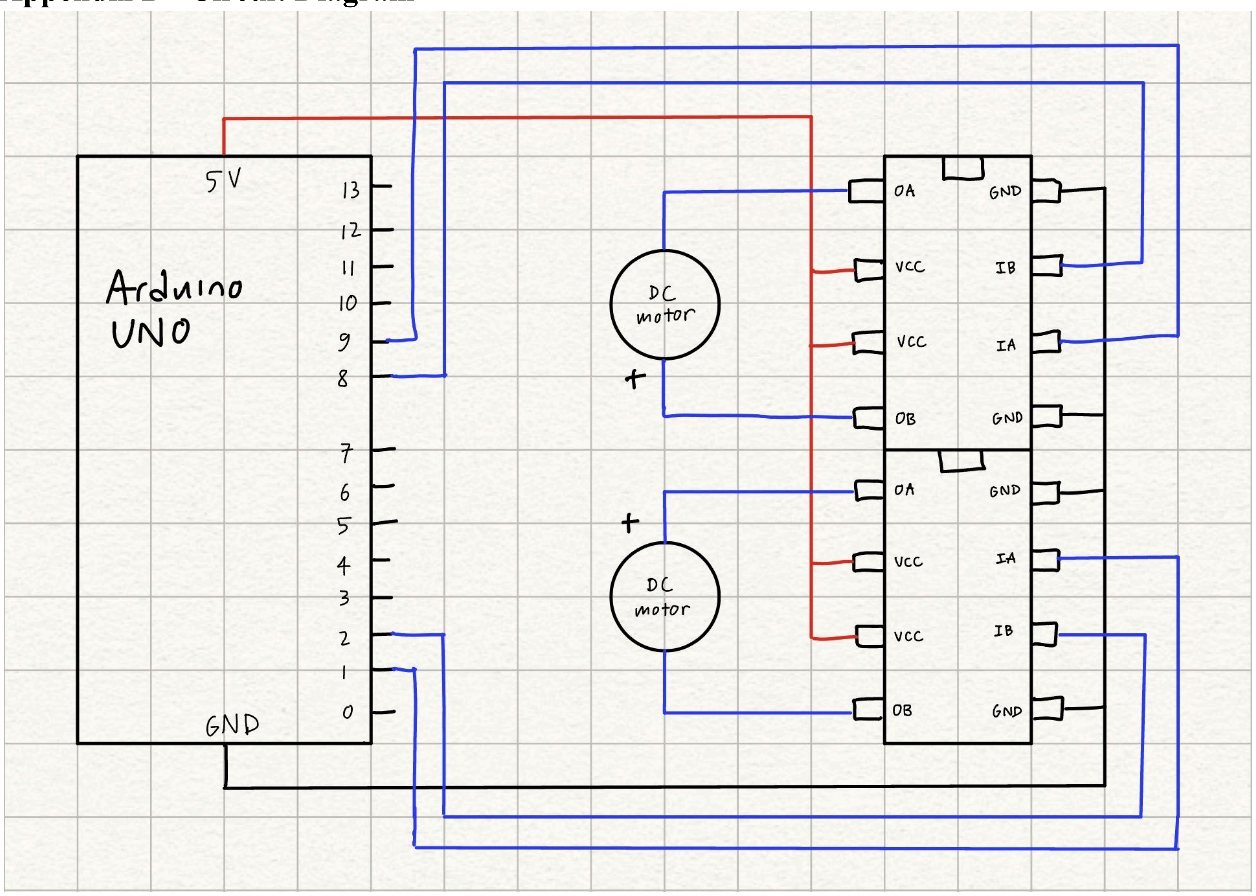
Conclusions

If we could change anything about our process, we would focus more on the physical design of the robot earlier on in the project. Having a strong design concept early on would have allowed us to start ordering parts earlier and reorder later if needed. We think we also would have benefited from more research on specific parts and their compatibility with our robot. We planned on using bigger, sturdier wheels, but ended up sticking with the ones we were given because the ones we ordered did not fit on to our motors. Testing our robot on the competition board more would have also likely led to a better performance on competition day. Some additional advice for future groups would be to find at least one meeting time a week that everyone on the team is free, just to check in and make sure everyone is on the same page.

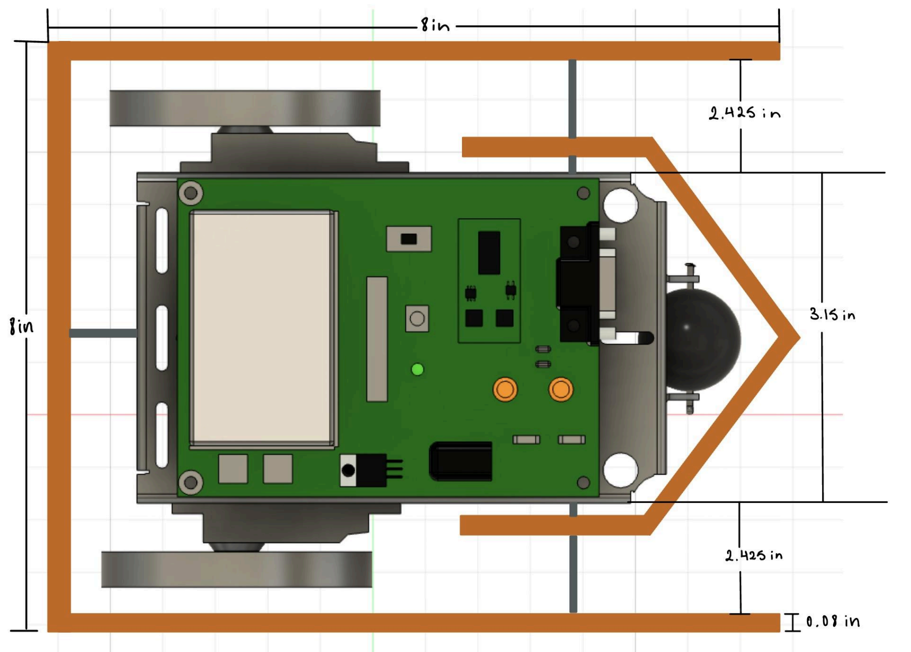
Appendix A - Bill of Materials

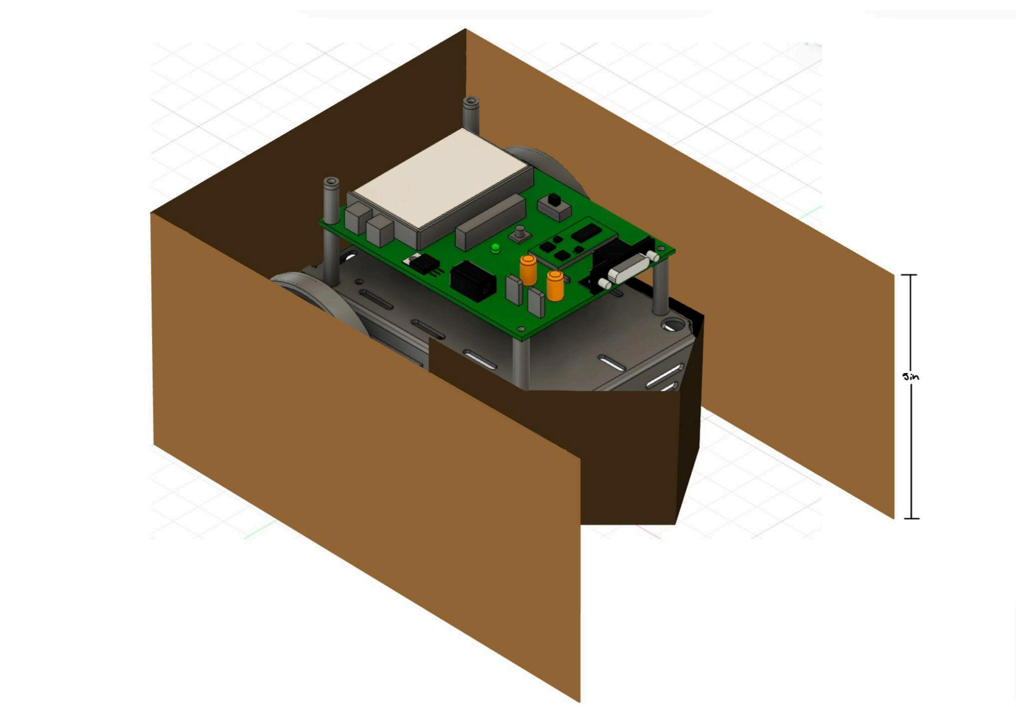
Part Name	Part #	Supplier	Unit Cost	Qty	Cost
Corrugated Sheet Plastic, 48" Wide x 48" Long	20585T28	McMaster-Carr	$26.64	1	$26.64
Black, Soft, 3" Diameter x 13/16" Wide	2337T41	McMaster-Carr	$2.44	2	$4.88
Cardboard box		Ella	$0	1	$0
				Total	$31.52

Appendix B - Circuit Diagram



Appendix C - CAD Files and Drawings





Although we never ended up using the wheels or corrugated plastic, here are the estimated volume and weight calculations for each wheel:

$V = \pi r^2 w = \pi(1.5 in)^2(0.813 in) = 5.7 in^3$

Using an approximate density of rubber, $\rho = 1.15 g/cm^3$,

$m = \rho V = (1.15\ g/cm^3)(93.4) cm^3 = 108 g$
$W = mg = 1N = 0.24\ lbforce$

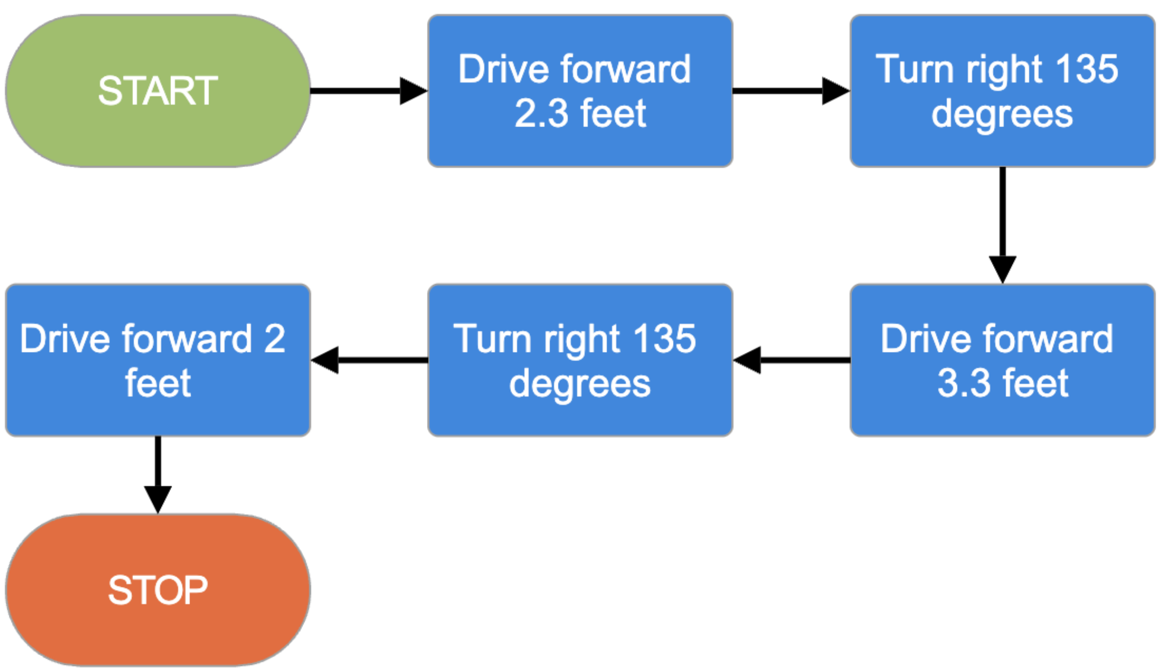
For the cardboard:

$V = 3(hwt) = 3(5\ in)(8\ in)(0.08\ in) = 9.6\ in^3$

Using an approximate density of cardboard, $\rho = 0.8 g/cm^3$,
$m = \rho V = (0.8\ g/cm^3)(157.3\ cm^3) = 125.8\ g$
$W = mg = 1.23\ N = 0.28\ lbforce$

The corrugated plastic never came in, so we were never able to dimension it.

Appendix D - Algorithm Flowchart



Appendix E - Arduino Code

```
/* FINAL CODE

/* NOTES
drive fwd - pins 2 and 8
  PORTD = 0b00000100; // pin 2
  PORTB = 0b00000001; // pin 8
turn right - pins 2 and 9
  PORTD = 0b00000100; // pin 2
  PORTB = 0b00000010; // pin 9
*/

int main(void){
  // set output pins
  DDRB = 0b00000011; // pins 8, 9 in port B
  DDRD = 0b00001100; // pins 2, 3 in port D

  // drive fwd 2.3 ft
  PORTD = 0b00000100; // pin 2
  PORTB = 0b00000001; // pin 8
  _delay_ms(3500);
  // reset pins
  PORTD = 0b00000000;
  PORTB = 0b00000000;
  _delay_ms(250);

  // turn right 135 degrees
  PORTD = 0b00000100; // pin 2
  PORTB = 0b00000010; // pin 9
  _delay_ms(750);
  // reset pins
  PORTD = 0b00000000;
  PORTB = 0b00000000;
  _delay_ms(250);

  // drive fwd 3.3 ft
  PORTD = 0b00000100; // pin 2
  PORTB = 0b00000001; // pin 8
  _delay_ms(5000);
  // reset pins
```

```
  PORTD = 0b00000000;
  PORTB = 0b00000000;
  _delay_ms(250);

  // turn right 135 degrees
  PORTD = 0b00000100; // pin 2
  PORTB = 0b00000010; // pin 9
  _delay_ms(750);
  // reset pins
  PORTD = 0b00000000;
  PORTB = 0b00000000;
  _delay_ms(250);

  // drive fwd 2 ft
  PORTD = 0b00000100; // pin 2
  PORTB = 0b00000001; // pin 8
  _delay_ms(3000);

  // stop
  PORTD = 0b00000000;
  PORTB = 0b00000000;
  _delay_ms(250);
}
```